

Correlation and Regression

Key terms

KSBs: K13, K54, S11, S52, B1

contingency table, explanatory and response variables, correlation, product moment correlation coefficient, covariance, regression, residual

So far, we have been studying univariate data, and looking at ways to analyse it with statistical methods. Often data is multivariate, and it is also useful to study *how two variables can be related.*

5.1 Contingency tables

If your data is *categorical* rather than numerical (e.g. counting the number of subjects who fall into particular categories, rather than measuring a numerical quantity for each subject), you can use a contingency table to study *the relationship between two variables.*

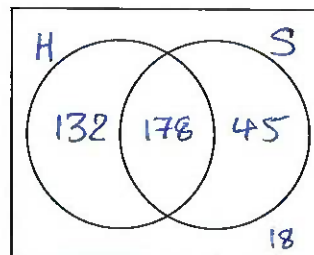
Definition 4.1 - Contingency Table

A **contingency table** (also called a **crosstab**, or a **two-way table**), can be used to summarise the relationship between categorical variables. *Two variables* can be shown simultaneously, with the quantities falling into each category shown in the appropriate *row/column of the table.*

Example 4.1. A survey has been conducted of *373 individuals* visiting a mountain lodge.

They were asked if they were there for hiking or skiing, and the results are shown in the table.

	Hiker	Non-hiker	Total
Skiier	178	45	<i>223</i>
Non-skiier	132	<i>18</i>	150
Total	<i>310</i>	63	373

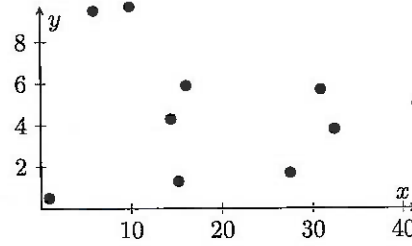
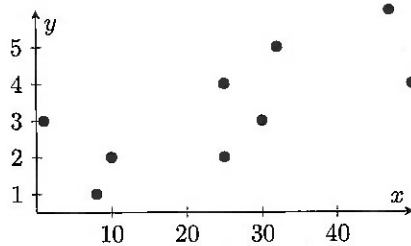


By looking at the table, we can see that *most people visiting the resort are there for hiking and more than half are there to ski* but there are a small number of people not participating in either activity.

Later we will see how statistical tests can be used to analyse the numbers in a contingency table.

4.2 Correlation

When the data we're measuring has multiple numerical variables, we can study how these variables are related to each other. The easiest way to see the relationship visually is on a *scatter plot*.

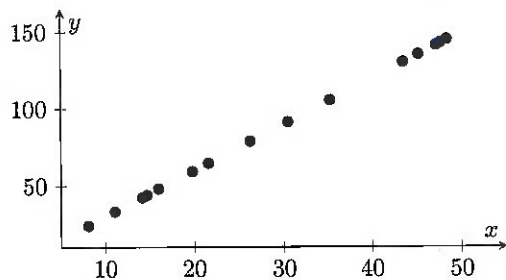


Definition 4.2 - Correlation

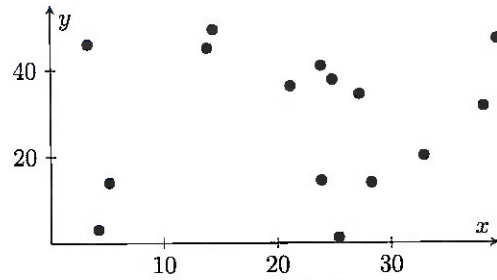
Correlation is a measure of the *relationship between two variables*.

- There are two types of correlation: *positive and negative*.
- A positive correlation is *when one variable increases the other one also increases*.
- A negative correlation is *when one variable increases the other one decreases*.
- If there isn't a relationship between two variables, we say *there is no correlation*.

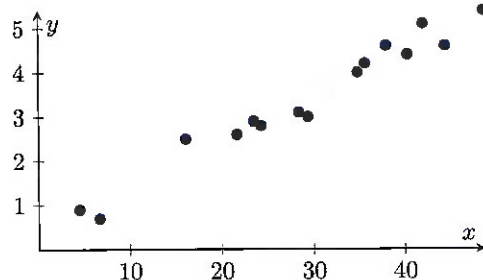
If the data were perfectly correlated - if they were related by a simple equation, like $y = 2x$, the data points would all lie *exactly on a straight line* (called a **perfect correlation**). When we have a plot which shows the points all being *quite close to the same line* we say the correlation is **strong**, and if the points are more spread out but *still following an overall direction* we say it is **weak**.



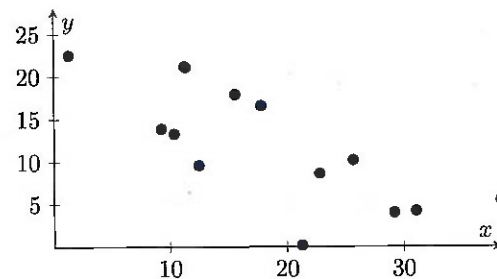
Perfect Positive Correlation



No Correlation



very strong positive correlation



Strong negative correlation

4.2.1 Covariance

There are a variety of mathematical techniques to *quantify correlation*, and one of them is covariance.

Definition 4.3 - Covariance

For a dataset with two variables X and Y , with means $\bar{x}$, $\bar{y}$ the **sample covariance** of X and Y , denoted $\text{cov}(X, Y)$ is given by

$$\text{cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{N - 1}$$

That is, we take the *difference between each variable and its mean* multiply them together, and take the mean of the result.

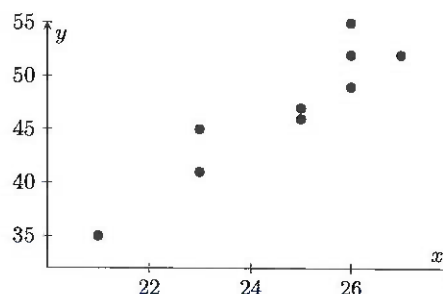
If the covariance is positive, this indicates the two variables have a positive correlation, and if the covariance is negative, this indicates the two variables have a negative correlation.

Note

There is also a population version of this, as for the variance, which involves dividing by *N instead of $N - 1$* .

Example 4.2. Say we measure the temperature each day in Eastbourne *for ten days in June*, and also count the number of *ice-creams sold each day*. We might expect to see data like this:

Temp ($^{\circ}\text{C}$) X	Ices sold Y	Temp ($^{\circ}\text{C}$) X	Ices sold Y
25	46	26	55
23	41	19	33
26	49	23	45
25	47	26	52
27	52	21	35



For each variable, we can calculate the mean:

$$\bar{x} = \frac{25 + 23 + \dots + 21}{10} = \frac{241}{10} = 24.1$$

$$\bar{y} = \frac{46 + 41 + \dots + 35}{10} = \frac{455}{10} = 45.5$$

Then the covariance is given by

$$\text{cov}(X, Y) = \frac{(25 - 24.1)(46 - 45.5) + (23 - 24.1)(41 - 45.5) + \dots}{9}$$

$$= \frac{(0.5 \times 0.5) + (-1.1 \times -4.5) + \dots}{9} = \frac{159.5}{9} = 17.72.$$

Since this is positive, it tells us *there is a positive correlation between these two variables*
 - when one increases, the other one also increases.

Note

The covariance of a variable with itself, $\text{cov}(X, X)$ is the *variance $\sigma^2(X)$* . The covariance is symmetric, so *$\text{cov}(X, Y) = \text{cov}(Y, X)$* .

The larger the size of the covariance is between two variables, the *stronger the correlation*.

Function: COVARIANCE.S (E/S)

The COVARIANCE.S() function (in Excel or Google Sheets) can be used to calculate the covariance of two sample data sets. It takes as an input two ranges of values (e.g. A1:A10, B1:B10).

Exercise 4.1. The data for the first two plots in this booklet is given in the tables below. Calculate the covariance for each, and comment on what it means.

X_1	Y_1
1	3
10	2
25	4
32	5
47	6
8	1
30	3
25	2
50	4

X_2	Y_2
32.4	3.8
0.9	0.5
30.9	5.7
14.4	4.3
5.9	9.5
41.7	5
15.2	1.3
27.5	1.7
9.9	9.7
16.1	5.9

4.2.2 Product Moment Correlation Coefficient

Another measure of correlation is the PMCC, also called *Pearson's correlation coefficient*.

Definition 4.4 - Product Moment Correlation Coefficient (PMCC)

For two variables X and Y , with standard deviations σ_x and σ_y the PMCC (denoted using the Greek letter ρ (rho)) can be calculated as:

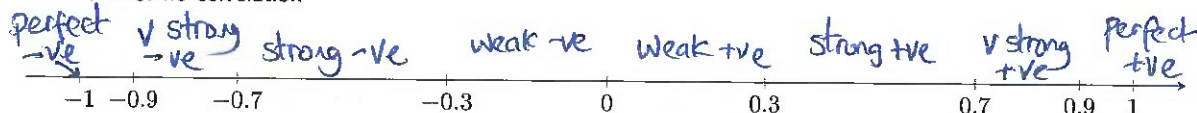
$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y}$$

That is, the *covariance of X and Y* divided by *the product of their S.D.s*.

Since we are scaling it down, the PMCC takes values *between -1 and 1*, and this allows us to compare correlations *across different datasets* and to quantify the *relative strength of a correlation*.

We can say that:

- $r = 1$ indicates a **perfect positive correlation.**
- $0.7 < r < 0.9$: very strong positive correlation
- $0.3 < r < 0.7$: strong positive correlation
- $0 < r < 0.3$: **weak positive correlation**
- $r = 0$: no correlation
- $-0.3 < r < 0$: weak negative correlation
- $-0.7 < r < -0.3$: strong negative correlation
- $-0.9 < r < -0.7$: **very strong negative correlation.**
- $r = -1$: perfect negative correlation



Example 4.3. For our previous ice-cream example (Example 4.2), the standard deviations of each dataset are $\sigma_x = 7.28$ and $\sigma_y = 2.56$. Then we can calculate the PMCC as:

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} = \frac{17.72}{7.28 \times 2.56} = 0.95.$$

Since this is quite close to (positive) 1, we can say that these two data sets have a **very strong positive correlation.**

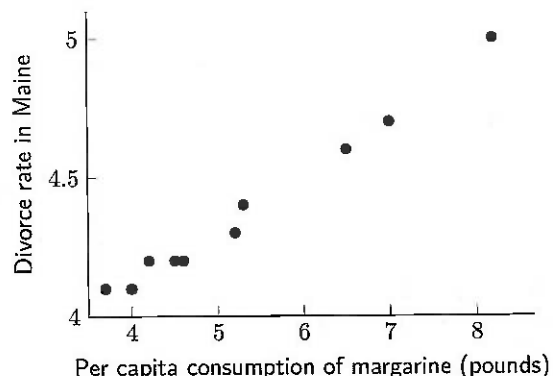
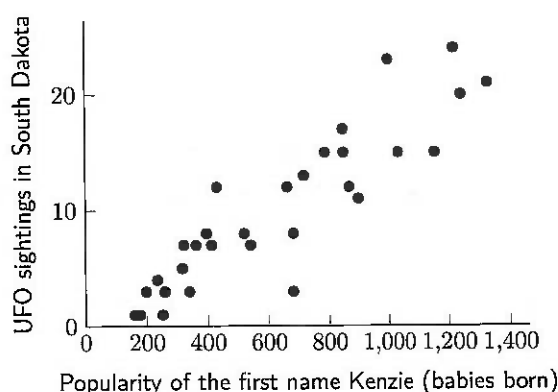
Exercise 4.2. For the two sets of points in Exercise 4.1, calculate the PMCC.

4.2.3 The meaning of correlation in data sets

If two data sets show a correlation, it is tempting to assume this is because of an **underlying causal link.** For example, we have seen a strong correlation between temperature and ice-cream sales (since when the weather is nice, it's likely that more people will be **out buying ice-cream**).

You might also expect that the rate of sunburn would be higher **when the temperature is higher.** But this would mean that if you plotted the incidence of **sunburn against ice cream sales**, they would likely display a **positive correlation.** This doesn't mean that you're **more likely to get sunburn if you have an ice cream:** the correlation is not because of causation, and there is a **third variable (temperature) affecting both**, which causes the correlation.

This means if you're studying a dataset and see a correlation between two variables, unless you understand **why they are related**, you can't make assumptions about why this correlation has occurred. One situation in which you can try to answer these questions is when one of the variables is **dependent on the other.**



4.3 Dependence in Bivariate Data

Sometimes, rather than having two random variables, one of the variables is *something we have control over* in taking measurements. This allows us to more closely observe the *relationship between the variables*, and to *fit a model to the data*.

Definition 4.5 - Independent and Dependent Variables

The **independent variable** (also called the **explanatory variable**, the **input variable** or the **control variable**) is a variable for which we can usually *control its value or choose to take readings* at certain points. Often, the independent variable is something like time or distance.

The **dependent variable** (also called the **response variable**, the **output variable** or the **measured variable**) is a variable whose value usually *depends on the value of the independent variable*. We can make observations of the value of the dependent variable, when the independent variable takes specific values, in order to *study the relationship*.

Note

The words 'dependent' and 'independent' here are being used with a slightly different meaning to the sense we use when we talk about independent and dependent events in probability, in which two events can be independent of each other, but one event on its own cannot be independent.

To study the relationship between variables, we can use **regression**, which involves *creating a model that fits the data* and using it to *make predictions*.

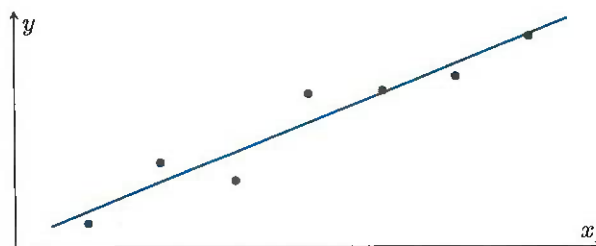
4.4 Regression

We might wish to express a model for the relationship between two variables more precisely. We could draw a line of best fit, also called a *regression line* that fits the data, and use it to make predictions.

Definition 4.6 - Regression Line

The **regression line** describes *the relationship* between the dependent variable and one or more independent variables in a set of data.

Roughly, the line looks like it runs through *the "middle" of the data*.

**Note**

We will mainly be working with data where there is one independent variable, but these methods can be extended to data with more variables.

4.4.1 Linear Regression

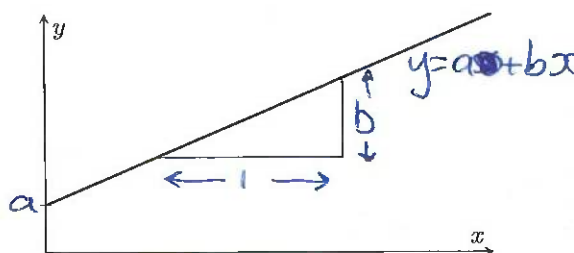
Many relationships in data are **linear**. This means that they can be described by *a straight line*. This usually happens when e.g. one variable is *a multiple of the other* and they increase at corresponding rates.

Many real-world examples follow a linear relationship: speed vs distance, house prices vs square footage, world records over time. This makes it an incredibly useful model for lots of situations, especially since linear plots are easy to *create, and visually interpret*.

We can define a straight line in terms of two parameters:

$$y = a + bx$$

This line has a slope of b (for every unit of distance the line *goes across, it goes up* b units) and a y -intercept of a (the value where it *crosses the y-axis*).



Exercise 4.3. Write down the equation for a line with each of the following properties, and then sketch the line.

- The line has a slope of 0.5 and a y -intercept of 1.5
- The line has a slope of -3 and a y -intercept of 4
- The line is horizontal and crosses the y -axis at 2
- The line starts at (0,1) and goes to the point (3,7)

The slope of the line tells us about the relationship between the two variables - if it is positive (the line slopes *upwards*), the variables are positively correlated, and if it is negative (the line slopes *downwards*), the variables have a negative correlation. However, it doesn't tell us much about the *strength of the correlation* - so you should use it in combination with the PMCC to get the full picture.

Definition 4.7 - Residual

For a data point (x, y) , the **residual** is the *vertical distance* between the point (x, y) and the *regression line* for the dataset.

Example 4.4.

This plot shows the ice-cream data from Example 4.2.

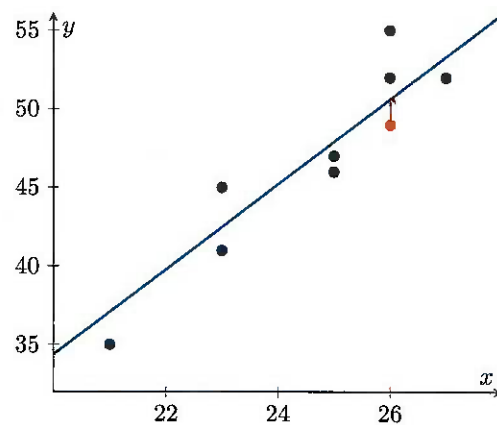
The regression line has equation $y = 2.71x - 19.76$. The marked data point has an x -value of *26* and a y -value of *49*.

The point directly above this on the regression line will have a y -value of

$$(2.71 \times 26) - 19.76 = 50.7$$

The residual is the difference between this value and its actual y -value, which is

$$50.7 - 49 = 1.7.$$



Exercise 4.4. For the regression lines and points given below, calculate the residual.

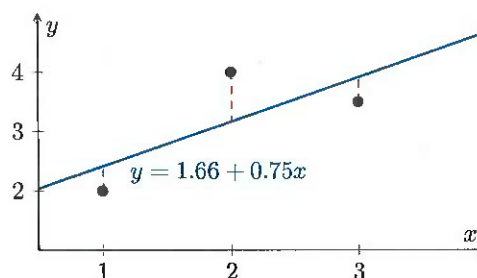
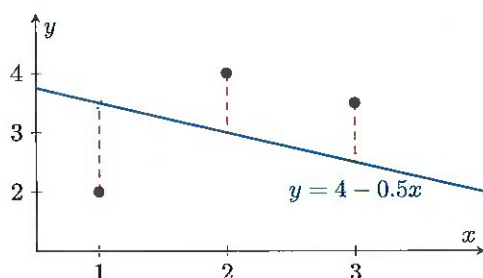
(a) $y = 5 + 2.8x$; $(-1.5, 2)$

(b) $y = 3.1 + 1.5x$; $(4, 1)$

(c) $y = 1 + 0.5x$; $(1.5, 3)$

4.4.2 Least Squares

The most common method used for linear regression is called **least squares regression**, and for a given set of data, it involves minimising the *squares of the residuals* of all the points in the set.



In the first plot, the residuals of the points are 1.5, 1 and 1. Taking the squares of these values and adding them together gives $1.5^2 + 1^2 + 1^2 = 4.25$. In the second plot, the sum of the squares of the residuals is $0.4166^2 + 0.8344^2 + 0.4166^2 = 1.04$. This is the smallest value it can take, and hence this is the *least squares regression line* for this data.

Finding this optimal line, in general, might be difficult from just looking at the data - but we can calculate the parameters for the line using a formula.

Definition 4.8 - Least Squares Regression Line

For a dataset with two variables X and Y , with means $\bar{x}$ and $\bar{y}$, the equation for the least squares regression line is given by $y = a + bx$, where

$$a = \bar{y} - b\bar{x} \quad \text{and} \quad b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

That is, for each data point in X , we calculate the *difference between it and the mean*, and multiply it by the equivalent value for the *matching data point in Y* ; we take the sum of all these values, and divide by the *sum of the squared of the distances to the mean* for values in X .

Once we have calculated b , we can use it, along with the means, to *find the value of a* .

Example 4.5. In the plots above, the three data points are $(1, 2)$, $(2, 4)$ and $(3, 3.5)$. The mean of the x values (1, 2 and 3) is 2, the mean of the y values (2, 4 and 3.5) is 3.166. Then

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{(1-2)(2-3.166) + (2-2)(4-3.166) + (3-2)(3.5-3.166)}{(1-2)^2 + (2-2)^2 + (3-2)^2}$$

$$= \frac{1.5}{2} = 0.75$$

$$a = \bar{y} - 0.75\bar{x} = 3.166 - (0.75 \times 2) = 1.66,$$

$$\Rightarrow y = 1.66 + 0.75x$$

Note

Note that the equation for the slope of the line is very similar to the equations for the **PMCC** and **covariance**. It's important to make sure you're using the right one! But this is not a surprise, since the slope of the regression line also indicates the direction of correlation between the variables.

Function: LINEST (E/S)

The `=LINEST()` function will calculate the slope b and y -intercept a of the least squares regression line for two sets of data. It takes three inputs:

- The range of y values
- The range of x values
- A TRUE/FALSE parameter for whether you would like the line to pass through $(0, 0)$; if the situation you're modelling will have $y = 0$ when $x = 0$, this is equivalent to adding $(0, 0)$ as an extra data point.
- There is also a fourth parameter to specify whether you want it to output a range of additional statistics for the data, including the standard error and residual sum.

The results of this formula will appear in the cell the formula is entered in (the value of b) and the one to the right (containing a). The statistics (if selected) will appear in a rectangle of cells below those.

Exercise 4.5. For the data sets given below, find the equation of the least squares regression line.

(a) $\begin{array}{c|c|c|c|c} X & 1 & 2 & 3 & 4 \\ \hline Y & 4 & 1 & 9 & 6 \end{array}$

$$\bar{x} = 2.5, \bar{y} = 5$$

(b) $\begin{array}{c|c|c|c|c} X & 9 & 9 & 4 & 4 \\ \hline Y & 9 & 10 & 7 & 4 \end{array}$

$$\bar{x} = 6.5, \bar{y} = 7.5$$

(c) $\begin{array}{c|c|c|c|c} X & 8 & 6 & 6 & 8 \\ \hline Y & 6 & 4 & 9 & 4 \end{array}$

$$\bar{x} = 7, \bar{y} = 5.75$$

For larger data structures with more variables, the least squares regression line is a line in **n -dimensional space**. It is harder to calculate for more variables, but we will see some approaches to this later.

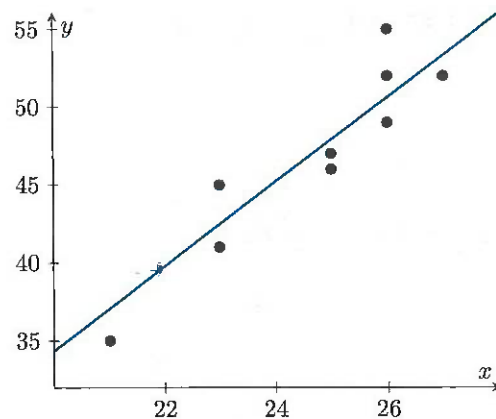
4.4.3 Using a Regression Line to make predictions

If we know the equation of the regression line, we can use it to *make predictions about data.*

Example 4.6. This is the ice-cream data from Example 4.2. The regression line has equation $y = 2.71x - 19.76$, and we would like to predict the amount of ice-cream we would sell for temperatures we don't have data about.

For example, if the temperature were 22°C ($x = 22$) we could predict that the number of ice-creams we'd sell would be:

$$\begin{aligned} y &= 2.71x - 19.76 \\ &= (2.71 \times 22) - 19.76 \\ &= 39.86. \end{aligned}$$

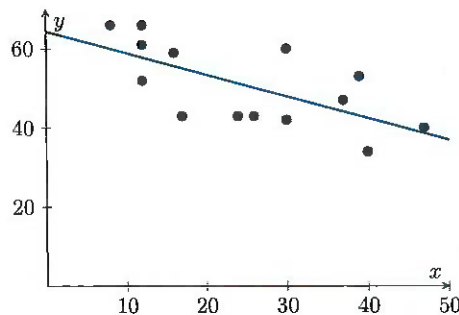
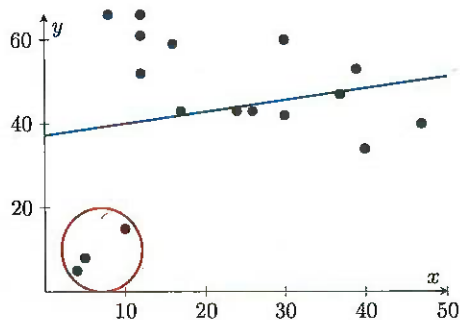


Exercise 4.6. For the datasets given in Exercise 4.5, predict what the output would be for the input value 5.

4.4.4 Outliers

If our dataset contains outliers, removing them can change the regression line significantly.

Example 4.7. In each of these plots, the regression line is shown, but the circled points in the left plot *have been removed*, and the regression line recalculated for the right plot.

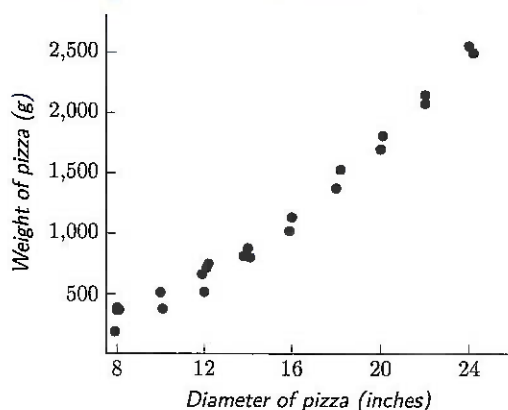


4.4.5 Nonlinear Relationships

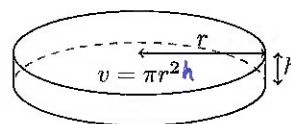
If your data does not have a linear relationship, this will be apparent from looking at a plot: instead of a straight line of points, there may instead be a *clearly defined curve*. Under these circumstances, using linear regression is *not a useful approach* as it will not provide a good model for how the data will behave.

Sometimes it will be possible to predict the type of relationship two variables will have, and if this is not linear, then linear regression will not be suitable.

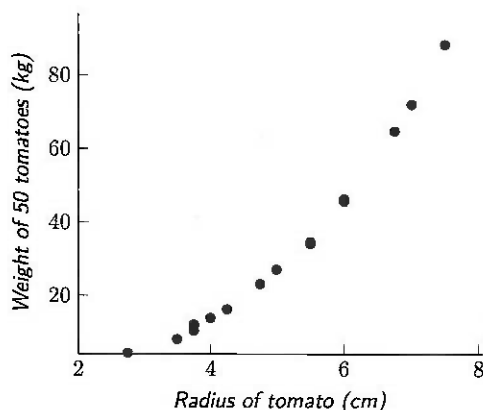
Example 4.8. In a pizza restaurant, the driver delivering pizzas wishes to estimate the amount of weight they will need to transport for different sizes of pizza. They measure a set of pizzas, and plot the *diameter against the weight of the pizza*.



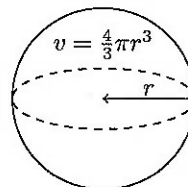
The weight of the pizza depends on *its volume* and for a fixed thickness, this can be described as $V = \pi r^2 \times h$ where h is the thickness of the pizza and r its radius. This will *not be a linear relationship* between v and r , since v depends on r^2 .



Later, the same driver is collecting vegetables from the supplier, and wishes to predict how much a box of 50 tomatoes will weigh. They plot the radius of an individual tomato against the weight of a box of 50 tomatoes of that size.



The weight of the box of tomatoes depends on *their volume*, and for a tomato of radius r this can be described as $V = \frac{4}{3}\pi r^3 \times 50$ since there are 50 tomatoes in the box. This will *not be a linear relationship* between v and r , since v depends on r^3 .



Even if a real-world situation doesn't have a direct linear relationship, the data can often be transformed to be presented *as a linear plot* - in the pizza example above, we could e.g. plot the weight against the *square of the radius* instead.